Year 1 AS Statistics and Mechanics

Chapter 9

Constant acceleration

Practice Booklet

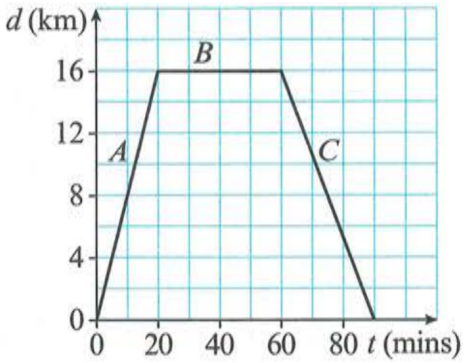
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9.1 Displacement–time graphs

Practice Book, printed page 58.

1. This is a displacement–time graph for a car travelling along a straight road. The journey is divided into three stages labelled A, B and C.



a. Work out the velocity of the car for each of the three stages of the journey, in km h^{-1}.

b. State the average velocity for the whole journey.

c. Work out the average speed for the whole journey.

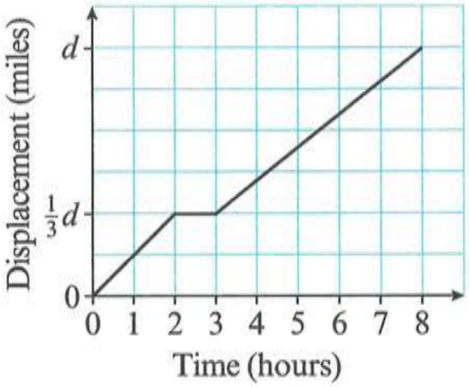
Hint

To find the velocity in km h^{-1}, convert the time from minutes to hours.

Hint

To find the average speed, divide the total distance travelled by the total time taken.

2. P This is a displacement–time graph for a train journey between Town A and Town C. The journey includes a stop to change trains at Town B. The three towns lie on a straight line.



During the first part of the journey, a third of the total distance, d, is covered in two hours. After a one-hour stop, the remaining distance is covered in five hours.
Given that the average velocity of the train between Town A and Town B is 65 mph, find:

a. the total distance of the journey, d

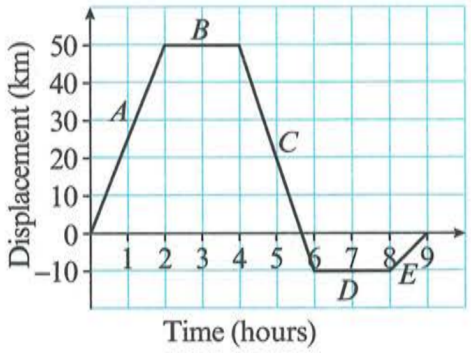
b. the average velocity of the train between Town B and Town C

c. the average velocity for the whole journey (including the stop).

Hint

To answer part **a**, remember that the gradient represents the velocity, so set up an equation and solve for d.

3. P Ken leaves home and cycles north in a straight line. He then travels back along the same route, continuing past his home. Later, he returns to his home.
The diagram shows a displacement–time graph for his journey divided into five stages, A–E.



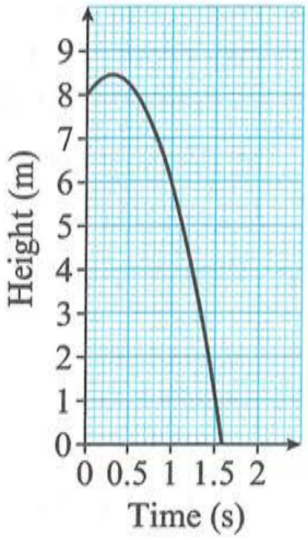
a. Work out the velocity for each stage of the journey.

b. Using the graph, find the length of time between Ken setting off, and passing his home again.

c. Using the graph, find the total length of time that Ken is more than 20 km from home.

d. Calculate the average speed for his entire journey.

4. P Rushanara jumps from a diving board into a swimming pool.
Rushanara is modelled as travelling in a straight vertical line. The displacement–time graph shows her vertical motion above the pool.

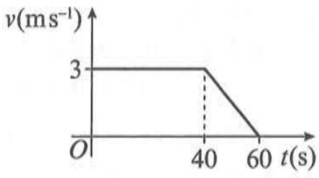


a. State the height of the diving board.
b. Using the graph, estimate Rushanara's maximum height above the pool and the time at which she reaches that height.
c. Write down Rushanara's velocity when she reaches the highest point.
d. Describe the motion of Rushanara:
i. from the time she jumps until she reaches her highest point
ii. after reaching the highest point.

9.2 Velocity–time graphs

Practice Book, printed page 59.

1. The diagram shows a velocity–time graph illustrating the motion of a runner moving along a straight road for a period of 60 seconds.



For the first 40 seconds, the runner moves at a constant speed of 3 m s^{-1}. The runner then decelerates at a constant rate, stopping after a further 20 seconds.

a. Find the displacement of the runner from the starting point after this 60-second period.
b. Work out the rate at which the runner decelerates.

Hint

For part **b**, the gradient represents the acceleration.

2. A car accelerates uniformly from rest at point A to a velocity of 18 m s^{-1} in 8 seconds. This velocity is then maintained for a further 12 seconds until the car passes point B.

a. Sketch a velocity–time graph to illustrate the motion of the car.

b. Find the rate at which the car accelerates.

c. Find the distance from A to B.

Hint

For part **c**, the area between a velocity–time graph and the horizontal axis represents the distance travelled.

3. P A cyclist is moving along a straight road. The cyclist starts from rest at point A and accelerates uniformly at a constant rate of 0.6 m s^{-2} for 20 seconds until reaching point B. The cyclist maintains a constant velocity for 30 seconds until reaching point C. The cyclist then decelerates uniformly for 10 seconds, coming to rest at point D.

a. Sketch a velocity–time graph to illustrate the motion of the cyclist.

b. Find the deceleration of the cyclist during the journey from C to D.

c. Find the distance from A to D.

d. Find the cyclist's displacement from the starting point after 30 seconds.

4. E/P A particle moves along a straight line. The particle accelerates uniformly from rest to a velocity of 10 m s^{-1} in T seconds. The particle then travels at a constant velocity of 10 m s^{-1} for 50 seconds. The particle then decelerates uniformly to rest in a further $3T$ seconds.

a. Sketch a velocity–time graph to illustrate the motion of the particle. (3 marks)

Given that the total displacement of the particle is 1000 m,

b. find the value of T. (3 marks)

5. E/P A particle is travelling in a straight line with velocity u m s^{-1}. At point A the particle accelerates at a constant rate, reaching a velocity of 25 m s^{-1} after 10 seconds. The particle then immediately decelerates at a constant rate, coming to rest at point B after another 10 seconds.

a. Sketch a velocity–time graph to illustrate the motion. (3 marks)

Given that the distance between A and B is 325 m,

b. find the value of u. (3 marks)

6. E/P Nishant runs a 100 m race in a straight line. He starts from rest and accelerates uniformly for 1 second, reaching a speed of 10 m s^{-1}. He maintains this speed for 5 seconds and then accelerates uniformly for 4 seconds, reaching a top speed of k m s^{-1}.

a. Sketch a velocity–time graph to illustrate the motion. (3 marks)

Nishant runs 100 m in 10 seconds.

b. Find the value of k. (3 marks)

c. Work out the acceleration of Nishant during the last 4 seconds of the race. (2 marks)

d. Suggest one improvement that could be made to the model of the motion of the sprinter during the race in order to make the model more realistic. (1 mark)

9.3 Constant acceleration formulae 1

Practice Book, printed page 60.

1. A particle is moving in a straight line with constant acceleration 5 m s^{-2}. Initially, the velocity of the particle is 3 m s^{-1}. Find the velocity of the particle after 10 seconds.

Hint

You need v and you know u, a and t so you can use $v = u + at$.

2. A cyclist decelerates uniformly while travelling on a straight road. The cyclist is initially travelling with velocity 9 m s^{-1} and comes to rest in 20 seconds. Find the deceleration of the cyclist.

Hint

For each question, write down the values you know and the values you need to find. This will help you to decide which formula to use.

3. A particle is moving in a straight line with constant acceleration. At time $t = 0$, the velocity of the particle is 1.2 m s^{-1}. At time $t = 7$, the velocity of the particle is 6.8 m s^{-1}. Find the distance travelled in these 7 seconds.

Hint

You need s and you know u, v and t, so use $s = \left(\frac{u+v}{2}\right) t$.

4. A car decelerates uniformly while travelling on a straight road. The car passes two signposts that are 720 m apart. The car passes the first signpost with velocity 26 m s^{-1}, and the second signpost with velocity 10 m s^{-1}. Find the time taken to travel between the two signposts.

5. A car decelerates uniformly at a rate of 2.4 m s^{-2} while travelling on a straight road. The car passes two signposts, taking 8 seconds to travel between them. As it passes the second signpost, the car has velocity 1.8 m s^{-1}. Find the velocity of the car as it passes the first signpost.

6. A particle moves in a straight line from a point A to a point B with constant acceleration. The particle takes 15 seconds to travel 120 m from A to B. The velocity of the particle at point A is 4.5 m s^{-1}. Find the velocity of the particle at point B.

7. A particle is moving in a straight line with constant acceleration 4 m s^{-2}. The velocity of the particle at A is 4 m s^{-1} and the velocity of the particle at B is 14 m s^{-1}. Find:

a. the time taken for the particle to move from A to B

b. the distance from A to B.

8. A particle moves in a straight line with constant acceleration from point A to point B. The particle takes 25 seconds to travel 400 m from A to B. The velocity of the particle at point B is 18 m s^{-1}. Find:

a. the velocity of the particle at point A

b. the acceleration of the particle.

9. A car accelerates uniformly while travelling on a straight road. The car is initially at rest and reaches a velocity of 15 m s^{-1} in 30 seconds. Find:

a. the acceleration of the car

b. the distance the car has travelled after 30 seconds

c. the velocity after 20 seconds

d. the distance the car has travelled after 20 seconds.

10. P A car travels 560 m in a straight line between points A and B. The velocity of the car at point A is 68.4 km h^{-1} and the velocity of the car at point B is 32.4 km h^{-1}. Find:

a. the time taken to travel from A to B

b. the deceleration of the car.

11. P Alex and Ben are cyclists racing on a straight 1 km road.
Alex accelerates uniformly from rest to a velocity of 8 m s^{-1} in 20 seconds. He then maintains the velocity of 8 m s^{-1} for the rest of the race.
Ben accelerates uniformly from rest to a velocity of 9 m s^{-1} in 50 seconds. He then maintains the velocity of 9 m s^{-1} for the rest of the race.
Both cyclists start at the same time. Determine who wins the race.

12. E/P A particle is moving in a straight line with constant deceleration between points A and B. The velocity of the particle at A is 22 m s^{-1}. The particle travels the 57 m from A to B in 3 seconds.

a. Show that the deceleration of the particle is 2 m s^{-2}. (4 marks)

After reaching B, the particle continues to move along the same straight line with the same constant deceleration. It comes to rest at point C.

b. Find the distance AC. (4 marks)

13. E/P A particle is moving in a straight line with constant acceleration a m s^{-2}.
The particle travels through points A, B and C at times $t = 0$, 5 and 15 seconds respectively.
The particle has velocity 15 m s^{-1} at point B and velocity 29 m s^{-1} at point C.

a. Show that the velocity of the particle at point A is 8 m s^{-1} and find the value of a. (4 marks)

b. Find the ratio of the distances $AB : BC$. (3 marks)

9.4 Constant acceleration formulae 2

Practice Book, printed page 62.

1. A particle is moving in a straight line with constant acceleration 3 m s^{-2}. It passes a point A with velocity 4 m s^{-1} and later passes through a point B, where $AB = 100$ m. Find the velocity of the particle as it passes through B.

Hint

You need v and you know u, a and s so you can use $v^2 = u^2 + 2as$.

2. A cyclist accelerates at a constant rate along a straight horizontal road. She starts from rest and reaches a velocity of 16 m s^{-1} after travelling 80 m. Find her acceleration.

Hint

Use $v^2 = u^2 + 2as$. Substitute the values of u, v and s and then solve the equation to find a.

3. A particle is moving in a straight line with constant acceleration 2.4 m s^{-2}. The particle's velocity is 2 m s^{-1} as it passes a point A, and 10 seconds later it passes a point B. Find the distance AB.

Hint

You need s and you know u, t and a, so use $s = ut + \frac{1}{2}at^2$. Always write down the formula you are using before you substitute any values.

4. A particle P is moving on the x-axis with constant deceleration. At time $t = 0$, the particle P passes through the origin O and is moving in the positive x-direction with speed 16 m s^{-1}. When $t = 8$, P is 32 m from O. Find the magnitude of the deceleration.

5. A car is decelerating at a constant rate of 1.5 m s^{-2} along a straight horizontal road between points A and B. The distance between points A and B is 75 m and the car has velocity 8 m s^{-1} when it passes point B. Find the velocity of the car at A.

6. A car is moving along a straight road with uniform acceleration 3.5 m s^{-2}. The car travels through a junction at speed 6 m s^{-1}. A second junction is 160 m further down the road. Find the time taken for the car to travel between the two junctions.

7. P A car approaches a stop sign at a constant speed of 12.5 m s^{-1}. When the car is 50 m from the stop sign, the driver applies the brakes with a force that causes the car to decelerate at a constant rate of 1.5 m s^{-2}. Determine whether or not the car will stop before the stop sign.

8. P A train is moving along a straight horizontal track with constant deceleration 0.25 m s^{-2}. The train passes a signal when travelling with velocity 54 km h^{-1}, and then passes a second signal at a velocity of 36 km h^{-1}. Find the distance between the signals.

9. E/P A car accelerates from rest to 30 m s^{-1} over a distance of 500 m.
Kieran models the car as travelling in a straight line with uniform acceleration.

a. Use this model to find the velocity of the car after it has travelled a distance of 200 m. (4 marks)

In fact, after it has travelled 200 m, the velocity of the car is found to be 16 m s^{-1}.

b. Comment on Kieran's model, and describe the acceleration of the car in light of this information. (1 mark)

10. E A particle moves along a straight line, from a point P to a point Q, with constant deceleration. The distance from P to Q is 450 m. The particle takes 20 seconds to move from P to Q and the speed of the particle at Q is 18 m s^{-1}. Find:

a. the deceleration of the particle (2 marks)

b. the speed of the particle at P. (2 marks)

11. E A particle moves along a straight line with constant acceleration 1.1 m s^{-2}. The particle moves 250 m in 19 seconds. Find to 1 decimal place:

a. the initial velocity of the particle (2 marks)

b. the final velocity of the particle. (2 marks)

12. E/P A motorcycle accelerates at a constant rate of 0.16 m s^{-2} along a straight horizontal road between points A and B which are 400 m apart. The motorcycle has velocity 18 m s^{-1} when it passes point B. Find the time taken to travel from A to B. (4 marks)

13. E/P A particle accelerates uniformly from point A to point B in 20 seconds. The velocities at A and B are 3 m s^{-1} and 28 m s^{-1} respectively. Given that M is the midpoint of AB, find the length of time it takes for the particle to travel from A to M. (6 marks)

14. E/P A particle moves along a straight line, with constant deceleration. It passes the points A, B and C at times $t = 0$, 4 and 10 respectively. Given that AB is 55.2 m and AC is 120 m:

a. find the initial velocity of the particle (5 marks)

b. find the time when the particle comes to rest. (4 marks)

9.5 Vertical motion under gravity

Practice Book, printed page 64.

1. A ball is dropped from a point 8 m above the ground with initial velocity 0 m s^{-1}. Find:

a. the speed of the ball after 1 second

b. the distance the ball has fallen after 1 second

c. the speed of the ball when it hits the ground

d. the time taken for the ball to hit the ground.

Hint

An object moving vertically under gravity can be modelled as a particle with a constant downward acceleration of $g = 9.8$ m s^{-2}.

2. A particle is thrown vertically downward from the top of a tower with speed 12 m s^{-1}. It reaches the ground in 2 seconds. Find:

a. the height of the tower

b. the speed of the particle when it hits the ground.

Hint

As the particle is moving downwards throughout its motion, it is sensible to take the downwards direction as positive.

3. A particle is thrown vertically downwards from the top of an 18 m tower.
The particle hits the ground after 1 second. Find:

a. the speed at which the particle was thrown from the tower

b. the speed of the particle when it hits the ground.

4. A ball is projected vertically upwards with speed 36 m s^{-1} from a point on the ground. Find:

a. the velocity of the ball after 3 seconds

b. the time of flight of the ball.

Hint

The time of flight is the total time that the ball is in motion; from the time that it is projected to the time that it stops moving. Here the ball will stop when it hits the ground, i.e. when $s = 0$.

5. A ball is projected vertically upwards with speed 11 m s^{-1} from a point on the ground. Find:

a. the greatest height above the point of projection reached by the ball

b. the time taken to reach this height.

6. A particle is thrown vertically upwards with speed u m s^{-1} from the ground. The particle reaches a maximum height of 10 m during its flight. Find:

a. the speed at which the particle was thrown

b. the time of flight of the particle.

7. E/P Isabel makes a dive from a high springboard into a diving pool. She leaves the springboard with speed 7 m s^{-1} upwards. When she leaves the springboard, she is 5 m above the surface of the pool. Isabel is modelled as a particle moving vertically under gravity alone and it is assumed that she does not hit the springboard as she descends.

a. Find the time taken for Isabel to reach the surface of the pool. (3 marks)

b. Find her speed when she reaches the surface of the pool. (3 marks)

In a refined model, the effect of air resistance is included. This refined model is now used to find the time taken by Isabel to reach the surface of the pool.

c. How would this new value of the time taken for Isabel to reach the surface of the pool compare with the value found using the initial model in part **a**? (1 mark)

8. E/P A particle is projected upwards from a point which is x m above the ground with speed 4.5 m s^{-1}. The time of flight for the particle is 2 seconds. Find the value of x. (4 marks)

9. E/P A particle is projected vertically upwards with speed 15 m s^{-1}. Find the total time for which the particle is 10 m or more above its point of projection. (4 marks)

10. E/P Pebble A is dropped from a tower 25 m high with initial velocity 0 m s^{-1}. One second later, pebble B is thrown downwards from the same tower with initial velocity u m s^{-1}. The two pebbles hit the ground at exactly the same time. Find the value of u to 3 significant figures. (5 marks)

11. E/P A ball is thrown vertically upwards from a point which is 10 m above the ground with speed u m s^{-1}. The ball hits the ground with speed $2u$ m s^{-1}. Find:

a. the value of u (5 marks)

b. the time of flight of the ball. (3 marks)

12. E/P A stone A falls vertically from rest from the top of a tower 90 m high. At the same time as A begins to fall, another stone B is projected vertically upwards from the bottom of the tower with speed 36 m s^{-1}. The stones collide. Find the time elapsed until the collision. (5 marks)

Problem solving: Set A

Bronze

A sprinter runs a straight 200 m race in T seconds. In a model of this race, it is assumed that, starting from rest, the sprinter accelerates uniformly for 6 seconds and reaches a speed of 6.6 m s^{-1}. The sprinter maintains this speed for the rest of the race.

a. Sketch a velocity–time graph for the motion of the sprinter during the whole race. (2 marks)

b. Find the acceleration of the sprinter during the first 6 seconds of the race. (1 mark)

c. Find the distance run by the sprinter in the first 6 seconds of the race. (1 mark)

d. Find the value of T to 3 significant figures. (3 marks)

Silver

Two particles A and B are moving in the same direction along the same straight horizontal road. At $t = 0$, A is at point O and is moving with velocity 20 m s^{-1}. A decelerates uniformly, and comes to rest at point X which is 400 m from O. At time $t = 0$, B is at point O and moves with constant velocity 12 m s^{-1}. From $t = T$ seconds, B decelerates uniformly and comes to rest at the same point X at the same instant as A does.

a. Sketch, on the same set of axes, the velocity–time graphs of the two particles for the period from $t = 0$ to the time when they both come to rest at the point X. (3 marks)

b. Find the value of T. (4 marks)

Gold

Two particles P and Q are moving in the same direction along the same straight horizontal road. At $t = 0$, P is at point O and is moving with constant velocity 8 m s^{-1}. From $t = T$ seconds, P decelerates uniformly, coming to rest at point X which is 450 m from O.
At time $t = 10$, Q is at point O. It is moving with velocity 10 m s^{-1} and decelerating uniformly. Q comes to rest at point X at the same instant as P.

a. By sketching the velocity–time graphs of the two particles, or otherwise, find the value of T. (4 marks)

b. Find the deceleration of each particle and show that the two particles never have the same instantaneous velocity before they come to rest. (5 marks)

Problem solving: Set B

Bronze

A ball is projected vertically upwards from a point which is x m above the ground. The ball has speed 25 m s^{-1} when it hits the ground. The time of flight of the ball is 4.5 seconds.
Find:

a. the speed at which the ball was projected (2 marks)

b. the value of x. (3 marks)

Silver

A ball is projected vertically upwards from a point that is 6 m above the ground. The ball reaches a maximum height of 13 m above the ground during its flight, before coming to rest on the ground. Find the total time of flight of the ball. (4 marks)

Gold

A particle is projected vertically upwards with a speed of 24.5 m s^{-1} from a point A.
The point B is h metres above A. The particle moves freely under gravity and is above B for 2 seconds. Calculate the value of h. (5 marks)

Selected exam question bank questions

Question 37

A car accelerates along a straight road. The car passes a road sign A, and then 30 seconds later passes a second road sign B. The distance from A to B is 750 m.

The car is modelled as a particle moving with constant acceleration.

Given that the speed of the car at B is 42 m s^{-1}, find:

a. the acceleration of the car (2 marks)

b. the speed of the car at A. (2 marks)

Question 39

A particle is travelling with velocity 20 m s^{-1}. At point A the particle accelerates at a constant rate, reaching a velocity of k m s^{-1} after 40 seconds. The particle then immediately decelerates a constant rate, coming to rest at point B after a further 20 seconds.

a. Sketch a velocity–time graph to illustrate the motion. (3 marks)

Given that the distance between A and B is 1120 m,

b. find the value of k. (3 marks)

Question 40

A particle is moving on the x-axis with constant deceleration 0.75 m s^{-2}. At time $t = 0$, the particle passes through the origin O, moving in the positive x-direction with speed 12 m s^{-1}. The particle reaches point A and then returns to O. Find:

a. the distance OA (2 marks)

b. the range of values for t for which the displacement from O is greater than 42 m. (4 marks)

Question 42

A car travels along a straight test track with constant acceleration. Three posts, P, Q and R, are fixed in that order at the side of the track.



The car passes post P travelling with a velocity of 6 m s^{-1}, then passes post Q one minute later with a velocity of 18 m s^{-1}.

a. Find:

i. the distance from P to Q

ii. the velocity of the car 30 seconds after it passes post P.

(5 marks)

Given that the ratio of the distances $PQ : QR = 3 : 2$,

b. find the time taken for the car to travel from Q to R. (3 marks)

Question 43

A car starts from rest at a signpost A and moves along a straight horizontal road. The car decelerates uniformly at a rate of 0.8 m s^{-2} from a speed of 21 m s^{-1} to a speed of 13 m s^{-1}. The speed is then maintained for 520 m. Finally, the car decelerates uniformly for 20 seconds before coming to rest at B.

a. For this journey from A to B, find:

i. the total time taken

ii. the distance from A to B.

(5 marks)

b. Sketch a displacement–time graph for the motion of the car, showing clearly the shape of the graph for each stage of the journey. (3 marks)

Question 45

A ball is projected upwards from a point which is x m above the ground with speed 12 m s^{-1}. The ball is modelled as a particle moving vertically under gravity. Given that the ball hits the ground 3 seconds after it is projected,

a. find the value of x. (4 marks)

In a refined model of the motion of the ball, the effect of air resistance is included. The speed of projection and the total time of flight are unchanged. This refined model is now used to find the value of x.

b. How would this new value of x compare with the value found using the initial model in part **a**? (1 mark)

Question 48

A ball is projected vertically upwards with speed 14 m s^{-1} from a point A, which is 3.6 m above the ground. Find:

a. the greatest height above A reached by the ball (3 marks)

b. the speed of the ball as it strikes the ground (3 marks)

c. the time between the instant when the ball is projected from A and the instant when the ball strikes the ground. (3 marks)

Question 51

A particle is moving in a straight horizontal line with constant deceleration 3 m s^{-2}. At time $t = 0$ the particle passes through a point O with speed 18 m s^{-1} and is travelling towards a point A, where $OA = 48$ m. Find:

a. the times when the particle passes through A (4 marks)

b. the value of t when the particle returns to O. (3 marks)

Question 52

A small ball is dropped from a height of 10 m onto level ground. It strikes the ground and bounces. The speed of the ball at the instant after it bounces is half its speed at the instant before it bounces. It rebounds to a height of x m, before falling again to the ground. The motion of the ball is modelled as that of a particle moving freely under gravity.

a.

i. Find the value of x

ii. Find the total time that elapses between when the ball is dropped and when it hits the ground for the second time.

(4 marks)

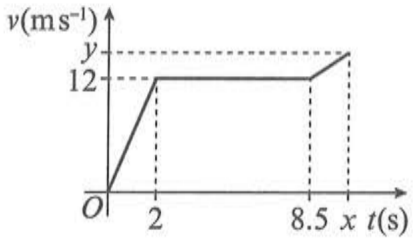
The model is refined to include air resistance, which is modelled as a constant resistive force of magnitude 0.5 N. Given that the mass of the ball is 0.2 kg,

b. find the speed of the ball immediately before it bounces (5 marks)

c. state, without calculation, how the value of x will differ to that found in part **a i** and give a reason for your answer. (1 mark)

Question 55

The velocity–time graph models the motion of a sprinter running a 100 m race. The sprinter accelerates uniformly from rest for 2 seconds, then travels at a constant speed of 12 m s^{-1} for 6.5 seconds, before accelerating uniformly to the finish line at a rate of 1.5 m s^{-2}.



a. Find, correct to 3 significant figures:

i. the total length of time the sprinter takes to finish the race, x seconds

ii. the speed of the sprinter as he crosses the finish line, y m s^{-1}.

(7 marks)

b. With reference to the velocity–time graph, give one criticism of the model of the sprinter's motion.

(1 mark)

Answers

9.1 Displacement–time graphs

1.

a. A: 48 km h^{-1}, B: 0 km h^{-1}, C: -32 km h^{-1}

b. 0 km h^{-1}

c. 21.3 km h^{-1} (3 s.f.)

2.

a. 390 miles

b. 52 mph

c. 48.75 mph

3.

a. A: 25 km h^{-1}, B: 0 km h^{-1}, C: -30 km h^{-1}, D: 0 km h^{-1}, E: 10 km h^{-1}

b. 5 hours 40 minutes

c. 4 hours 15 minutes

d. 13.3 km h^{-1}

4.

a. 8 m

b. 8.5 m, 0.3 s

c. 0 m s^{-1}

d.

i. She travels upwards and her speed decreases.

ii. She travels downwards and her speed increases.

9.2 Velocity–time graphs

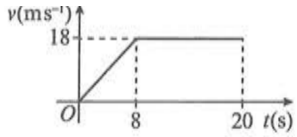
1.

a. 150 m

b. 0.15 m s^{-2}

2.

a.

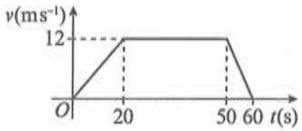


b. 2.25 m s^{-2}

c. 288 m

3.

a.



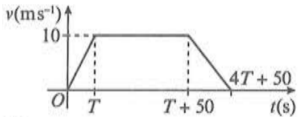
b. 1.2 m s^{-2}

c. 540 m

d. 240 m

4.

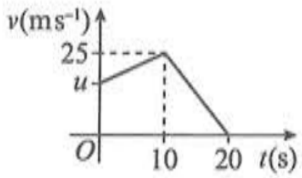
a.



b. 25

5.

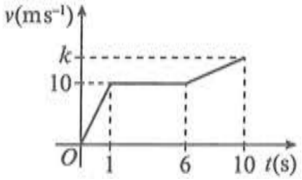
a.



b. 15

6.

a.



b. 12.5

c. 0.625 m s^{-2}

d. Rate of acceleration decreases as velocity increases.

9.3 Constant acceleration formulae 1

1. 53 m s^{-1}

2. 0.45 m s^{-2}

3. 28 m

4. 40 seconds

5. 21 m s^{-1}

6. 11.5 m s^{-1}

7.

a. 2.5 seconds

b. 22.5 m

8.

a. 14 m s^{-1}

b. 0.16 m s^{-2}

9.

a. 0.5 m s^{-2}

b. 225 m

c. 10 m s^{-1}

d. 100 m

10.

a. 40 seconds

b. 0.25 m s^{-2}

11. Alex

12.

a. $57 = \frac{22 + v}{2} \times 3 \Rightarrow v = 16$ m s^{-1}
$16 = 22 + a(3) \Rightarrow a = -2$ m s^{-2}

b. 121 m

13.

a. At $t = 5$, $15 = u + 5a$; at $t = 15$, $29 = u + 15a$.
So $3u - u = (3 \times 15) - 29 \Rightarrow u = 8$ m s^{-1}
$a = 1.4$ m s^{-2}

b. 57.5 : 220

9.4 Constant acceleration formulae 2

1. 24.8 m s^{-1} (1 d.p.)

2. 1.6 m s^{-2}

3. 140 m

4. 3 m s^{-2}

5. 17 m s^{-1}

6. 8 seconds

7. No, the car will not stop before the sign.

8. 250 m

9.

a. 19.0 m s^{-1} (1 d.p.)

b. Assumption of uniform acceleration seems to be incorrect. The car is accelerating at a lower rate in the first 200 m than in the final 300 m.

10.

a. 0.45 m s^{-2}

b. 27 m s^{-1}

11.

a. 2.7 m s^{-1} (1 d.p.)

b. 23.6 m s^{-1} (1 d.p.)

12. 25 seconds

13. 13.5 s (1 d.p.)

14.

a. 15 m s^{-1}

b. 25 seconds

9.5 Vertical motion under gravity

1.

a. 9.8 m s^{-1}

b. 4.9 m

c. 12.5 m s^{-1} (3 s.f.)

d. 1.28 seconds (3 s.f.)

2.

a. 43.6 m

b. 31.6 m s^{-1}

3.

a. 13.1 m s^{-1}

b. 22.9 m s^{-1}

4.

a. 6.6 m s^{-1}

b. 7.35 seconds (3 s.f.)

5.

a. 6.17 m (3 s.f.)

b. 1.12 seconds (3 s.f.)

6.

a. 14 m s^{-1}

b. 2.86 seconds (3 s.f.)

7.

a. 1.95 seconds (3 s.f.)

b. 12.1 m s^{-1} (3 s.f.)

c. New time will be greater than 1.95 seconds.

8. 10.6

9. 1.10 seconds (3 s.f.)

10. 13.7

11.

a. 8.08 (3 s.f.)

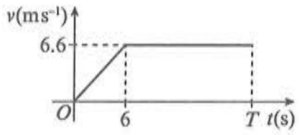
b. 2.47 seconds (3 s.f.)

12. 2.5 seconds

Problem solving: Set A

Bronze

a.



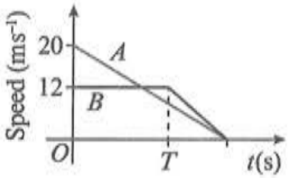
b. 1.1 m s^{-2}

c. 19.8 m

d. 33.3 (3 s.f.)

Silver

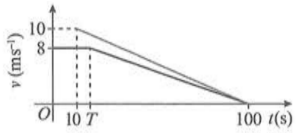
a.



b. 26.7 (3 s.f.)

Gold

a.



$T = 12.5$

b. Deceleration of $P = 0.0914$ m s^{-2} (3 s.f.)
Deceleration of $Q = 0.111$ m s^{-2} (3 s.f.)
As rate of deceleration of Q is greater than the rate of deceleration of P the particles will only have the same instantaneous velocity if velocity of $Q = 8$ for some t where $10 < t < 12.5$.
Velocity of $Q = 10 + (-0.111)(t - 10)$
$10 + (-0.111)(t - 10) = 8 \Rightarrow t = 28 > 12.5$
The particles never have the same instantaneous velocity.

Problem solving: Set B

Bronze

a. 19.1 m s^{-1}

b. 13.275

Silver 2.82 seconds (3 s.f.)

Gold 25.7 m (3 s.f.)

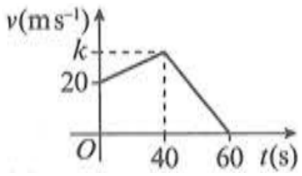
Selected exam question bank answers

Question 37

a. 1.13 m s^{-2} (3 s.f.)
b. 8 m s^{-1}

Question 39

a.



b. 24

Question 40

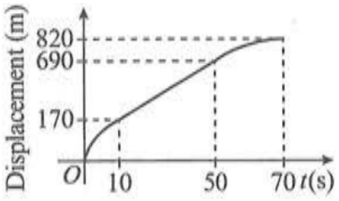
a. 96 m
b. $4 < t < 28$

Question 42

a.
i. 720 m
ii. 12 m s^{-1}
b. 23.6 seconds

Question 43

a.
i. 70 seconds
ii. 820 m
b.



Question 45

a. 8.1 m
b. New value is lower

Question 48

a. 10 m
b. 16.3 m s^{-1} (3 s.f.)
c. 3.09 seconds (3 s.f.)

Question 51

a. 4 and 8 seconds

b. 12

Question 52

a.

i. 2.5

ii. 2.86 seconds (3 s.f.)

b. 12.1 m s^{-1} (3 s.f.)

c. x will be smaller, because the speed after rebound will be lower.

Question 55

a.

i. 9.29 s (3 s.f.)

ii. 13.2 m s^{-1} (3 s.f.)

b. For example, it is impossible for the acceleration of the sprinter to change instantaneously, so the graph could not have 'sharp corners'